

POWER BI

From Beginner to Advanced | Concept-Wise Structured Guide
Data Analytics & Business Intelligence

Curriculum



POWER BI

From Beginner to Advanced | Concept-Wise Structured Guide Data Analytics & Business Intelligence

Module 1: Introduction to Power BI

Concept 1.1 — What is Power BI?

- Overview of Business Intelligence (BI) and data analytics
- Power BI as a self-service BI tool by Microsoft
- Key components: Power BI Desktop, Power BI Service, Power BI Mobile
- Use cases: dashboards, reports, KPI tracking, data storytelling

Concept 1.2 — Power BI Architecture

- The ETL Pipeline: Extract, Transform, Load
- Data sources → Power Query → Data Model → Reports → Dashboards
- On-premises vs cloud data connectivity
- Difference between reports and dashboards

Concept 1.3 — Installation & Interface

- Downloading and installing Power BI Desktop
- Navigating the Power BI Desktop interface
- Report View, Data View, Model View
- The ribbon, fields pane, filters pane, visualizations pane

Concept 1.4 — Power BI Workflow

- Connect to data source
- Transform and clean data with Power Query
- Build data model (relationships)
- Create visualizations and apply DAX measures
- Publish to Power BI Service

Activity	Description
Practical Exercise	Build your first report using a sample Excel dataset
Key Deliverable	A simple 1-page Power BI report with 3 visuals

Module 2: Connecting to Data Sources

Concept 2.1 — File-Based Sources

- Connecting to Excel workbooks and CSV files
- Importing JSON and XML files
- Folder connector: combining multiple files
- PDF data extraction

Concept 2.2 — Database Sources

- SQL Server, MySQL, PostgreSQL connections
- Import mode vs DirectQuery mode vs Live Connection
- Stored procedures and native queries
- Authentication methods (Windows, Database, OAuth)

Concept 2.3 — Cloud & Online Sources

- SharePoint Online Lists and Folders
- Azure SQL Database and Azure Data Lake
- Connecting to Google Analytics and Salesforce
- Web connector: scraping HTML tables
- REST API connection basics

Concept 2.4 — Import vs DirectQuery

Mode	How it Works	Advantage	Limitation
Import Mode	Data loaded into memory	Faster performance	Scheduled refresh needed
DirectQuery	Queries run live on source	Always current data	Slower on large datasets
Live Connection	SSAS / Power BI Datasets	Shared models	Limited transformations

Module 3: Power Query — Data Transformation

Concept 3.1 — Power Query Editor Interface

- Launching Power Query Editor
- Applied Steps panel and query settings
- Data type management and column profiling
- Query dependencies and advanced editor

Concept 3.2 — Data Cleaning Essentials

- Remove duplicates, errors, and blank rows
- Replace values and fill up/down
- Trim, clean, and format text columns
- Change data types (text, number, date, boolean)

Concept 3.3 — Column Transformations

- Split columns by delimiter or character count
- Merge columns and conditional column creation
- Extract: first N chars, last N chars, text between
- Custom columns using M expressions

Concept 3.4 — Table Transformations

- Transpose, pivot, and unpivot columns
- Group By: aggregations on grouped data
- Append queries: stacking tables vertically
- Merge queries: joins (inner, left, full outer)

Concept 3.5 — M Language Basics

- Understanding M formula language syntax
- let...in expression structure
- Common M functions: Text.Upper, Date.Year, List.Sum
- Parameters and parameterised queries

Module 4: Data Modeling

Concept 4.1 — Data Modeling Fundamentals

- What is a data model and why it matters
- Fact tables vs dimension tables
- Star schema vs snowflake schema
- Flat tables: limitations and when to avoid them

Concept 4.2 — Relationships

- Creating relationships between tables
- Cardinality: One-to-Many, Many-to-One, One-to-One, Many-to-Many
- Cross filter direction: Single vs Both
- Active vs Inactive relationships
- USERRELATIONSHIP() to activate inactive relationships

Concept 4.3 — Date Table (Calendar Table)

- Importance of a dedicated date dimension
- Creating a date table using DAX CALENDAR() and CALENDARAUTO()
- Adding Year, Quarter, Month, Week columns
- Marking a table as Date Table
- Connecting date table to fact tables

Concept 4.4 — Model Optimization

- Hiding unnecessary columns from report view
- Setting sort-by-column (e.g., sort Month Name by Month Number)
- Data categories: URL, image URL, geography
- Hierarchies: Year > Quarter > Month > Day

Module 5: DAX — Data Analysis Expressions

Concept 5.1 — DAX Fundamentals

- Measures vs Calculated Columns vs Calculated Tables
- DAX syntax: table[column] and measure names

- Context: Row context vs Filter context
- Implicit vs explicit measures

Concept 5.2 — Basic DAX Functions

Category	Key Functions
Aggregation	SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, DISTINCTCOUNT
Logical	IF, AND, OR, NOT, SWITCH, IFERROR
Text	CONCATENATE, LEFT, RIGHT, MID, UPPER, LOWER, FORMAT
Date & Time	TODAY, NOW, YEAR, MONTH, DAY, DATEDIFF, EOMONTH
Math	DIVIDE, ROUND, ABS, INT, MOD, POWER, SQRT

Concept 5.3 — Filter Functions

- CALCULATE: the most important DAX function
- ALL, ALLEXCEPT, REMOVEFILTERS: removing filters
- FILTER: row-by-row filtering with conditions
- RELATED and RELATEDTABLE: cross-table lookups
- SELECTEDVALUE and HASONEVALUE

Concept 5.4 — Time Intelligence

- TOTALYTD, TOTALQTD, TOTALMTD: period-to-date calculations
- DATEADD: shift dates by periods (e.g., prior year)
- SAMEPERIODLASTYEAR: year-over-year comparison
- DATESBETWEEN and DATESINPERIOD
- Rolling averages and running totals

Concept 5.5 — Advanced DAX

- SUMX, AVERAGEX, MAXX, MINX: iterator (X) functions
- RANKX: ranking with dynamic filters
- TOPN: selecting top N rows
- VAR and RETURN: variables in DAX
- EARLIER and EARLIEST: row context in iterators

Module 6: Visualizations & Report Design

Concept 6.1 — Core Visuals

Visual Type	Best Used For	Example Use Case
Bar / Column Chart	Category comparisons	Sales by region, Product ranking
Line Chart	Trends over time	Monthly revenue trend
Pie / Donut Chart	Part-to-whole (max 5 slices)	Market share by category
Card & Multi-row Card	Single KPI display	Total Sales, YTD Revenue
Table & Matrix	Detailed tabular data	Sales pivot by region & product
Map / Filled Map	Geographic distribution	Sales by country/state
Scatter Chart	Correlation analysis	Sales vs Profit by product
Treemap	Hierarchical part-to-whole	Revenue by category and subcategory
Gauge	Progress toward target	Sales vs goal
Waterfall Chart	Running total changes	Profit bridge analysis

Concept 6.2 — Interactivity

- Slicers: filtering reports by dimension
- Cross-filtering and cross-highlighting between visuals
- Drill-down and drill-through navigation
- Bookmarks: saving report states and creating navigation
- Tooltips: default, report page, and custom tooltips
- Buttons and actions for navigation

Concept 6.3 — Formatting & Themes

- Applying JSON themes for brand consistency
- Conditional formatting: background, font, data bars, icons
- Dynamic titles using DAX SELECTEDVALUE()
- Canvas background, wallpaper, and layout design
- Report design principles: alignment, contrast, hierarchy

Concept 6.4 — Custom Visuals

- Importing visuals from AppSource
- Deneb: Vega-Lite custom visuals
- HTML Content visual for rich text
- Using R and Python visuals

Module 7: Power BI Service & Deployment

Concept 7.1 — Power BI Service Overview

- Workspaces: My Workspace vs shared workspaces
- Publishing reports from Desktop to Service
- Apps: packaging reports for end-user distribution
- Licensing: Free, Pro, and Premium Per User

Concept 7.2 — Data Refresh

- Scheduled refresh setup and frequency
- On-premises Data Gateway: purpose and configuration
- Incremental refresh for large datasets
- Refresh history and failure troubleshooting

Concept 7.3 — Collaboration & Sharing

- Sharing reports and dashboards with individuals
- Row-Level Security (RLS): static and dynamic
- Workspace roles: Admin, Member, Contributor, Viewer
- Subscriptions and email alerts
- Embedding reports in SharePoint and Teams

Concept 7.4 — Dataflows & Datasets

- Dataflows: reusable Power Query transformations in the cloud
- Certified and promoted datasets
- Shared datasets and live connections
- Data lineage view

Module 8: Advanced Topics & Best Practices

Concept 8.1 — Performance Optimization

- Performance Analyzer: identifying slow visuals
- Reducing model size: removing unused columns
- Aggregations and composite models
- Avoiding bi-directional relationships
- Using integer keys instead of text for relationships

Concept 8.2 — Advanced Analytics

- Quick Insights and AI Insights in Power BI Service
- Key Influencers visual: automated ML insights
- Decomposition Tree: drill-down analysis
- Anomaly Detection in line charts
- Smart Narratives: auto-generated summaries
- Q&A visual: natural language queries

Concept 8.3 — Power BI & Python/R

- Running Python scripts for data transformation
- Python-based visualizations (matplotlib, seaborn)
- R scripts in Power Query and visuals
- Machine learning model integration via Azure ML

Concept 8.4 — Enterprise Best Practices

- Naming conventions for measures, tables, and columns
- Documentation and report annotations
- Version control with Tabular Editor and Git
- XMLA endpoint and ALM Toolkit
- Deployment pipelines: Dev → Test → Production

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